

Yulong Liu

Ph.D. Candidate · Cornell University

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Research Focus

Differentiable programming, hybrid neural PDE solvers, and uncertainty-aware scientific AI for multiphysics geomechanical systems.

Education

Cornell University

Ph.D. in Earth Science; Department of Earth and Atmospheric Sciences

Advisor: Chloé Arson. Minors: Computer Science and Scientific Computing.

2024 – Present

Ithaca, NY

Northeastern University

B.S. in Mining Engineering

2020 – 2024

Shenyang, China

Research Experience

Doctoral Research | Cornell University

Aug. 2024 – Present

- Develop coupled thermo-hydro-mechanical models in the MOOSE finite element framework for fractured rock and geothermal injection–production systems.
- Study differentiable programming and hybrid neural PDE solvers, combining physics-informed networks, implicit neural representations, and neural operators with numerical simulation.
- Investigate uncertainty-aware surrogate modeling for geothermal systems, with an emphasis on uncertainty propagation and the reliability of multiphysics predictions.
- Lead a cross-institution project on PDE-grounded verification of the physical correctness of LLM-generated multiphysics simulation code.
- Study multiscale transport and AI-assisted discovery of interpretable permeability laws using homogenization theory for heterogeneous porous media.

Undergraduate Research | Northeastern University

Sep. 2020 – Jun. 2024

- Thesis: *A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning*. Developed a joint deep-learning model; received the Outstanding Thesis Award (2024).
- Conducted experimental and computational work on hydraulic fracturing, triaxial rock testing, and tunnel boring machines.

Grants, Proposals & Computational Awards

NSF ACCESS | 1.5M CPU hours

2026

Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs

Student lead: Y. Liu; advisor: C. Arson. Total allocation and Arson group share: 1.5M CPU hours.

Empire AI Alpha+Beta | AI-aided Computational Geomechanics

2026

PIs: Y. C. Han, Y. Liu, A. Tristani, C. Arson. Total allocation and Arson group share: 5.5K GPU hours.

NASA FINESST | Proposal under review

2026

ROSES-25 F.5. Future Investigator: Y. Liu; PI: C. Arson. Title withheld during anonymous review.

Technical Expertise

Computing

Python, C++, R; MOOSE; finite element and coupled multiphysics simulation.

Scientific AI

Physics-informed neural networks, neural operators, implicit neural representations; PDE-grounded verification of AI-generated simulation code.

Geomechanics

Poromechanics, hydraulic fracturing, geothermal reservoir modeling, multiscale transport and homogenization; triaxial rock testing and laboratory instrumentation.

Tools

AutoCAD, SolidWorks, L^AT_EX/Overleaf. Continuing study: phase-field modeling and multiphase flow in porous media.

Publications

Journal articles

1. **Yulong Liu**, Chloé Arson. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock*. Rock Mechanics and Rock Engineering, 2026. DOI: 10.1007/s00603-026-05882-5.
2. M. Belachew, **Yulong Liu**, J. D. Frost, Chloé Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling*. Tunnelling and Underground Space Technology, 172, 107501, 2026. DOI: 10.1016/j.tust.2026.107501.

Conference papers

3. **Yulong Liu**, Jonah Botvinick-Greenhouse, Yunan Yang, Chloé Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study*. ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
4. **Yulong Liu**[†], Chloé Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media*. ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.

Preprints & manuscripts

5. **Yulong Liu**^{*,†}, Zhenghan Song, et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code*. arXiv:2605.09360. Under review at NeurIPS 2026.
6. Zhenghan Song, Yunyi Li, **Yulong Liu**[†]. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking*. Under review at ACL ARR 2026.

* Co-first author † Corresponding author

Talks & Presentations

Oral presentation , 60th US Rock Mechanics/Geomechanics Symposium (ARMA) Tucson, AZ, USA	Jun. 2026
Invited talk , 20th Phase Field Methods Workshop Center for Hierarchical Materials Design, Northwestern University, Evanston, IL	May 2026
Poster , Cornell CEE Graduate Research Symposium Cornell University, Ithaca, NY	Apr. 2026
Invited seminar , Physics-Informed AI Optimization of Heterogeneous Rock Systems Southwest Petroleum University, Chengdu, China	Dec. 2025
Poster , 59th US Rock Mechanics/Geomechanics Symposium (ARMA) Santa Fe, NM, USA	Jun. 2025

Honors & Awards

Estwing Hammer Award , Cornell University Most Outstanding EAS Graduate Student of 2024	2025
Cornell University Travel Grant	2025, 2026
Outstanding Thesis , Northeastern University	2024
National Scholarship , Ministry of Education, China	2024
First Class Scholarship , Northeastern University	2023
Autumn Scholarship & Golden Seed Scholarship , Northeastern University	2022

Academic Service & Memberships

Reviewer: Rock Mechanics and Rock Engineering; Underground Space; International Conference on Artificial Neural Networks; Geothermal Rising Conference.

Memberships: American Rock Mechanics Association (2025–present); International Society for Rock Mechanics and Rock Engineering (2025–present); Chinese Society for Rock Mechanics and Engineering (2024–present); Geothermal Rising (2025–present).